

STELLARHACKATHON // MISSION DOSSIER

AETHERION

Project ASTRA — Adaptive Stellar Threat Response Array

A COLONY NETWORK ORBITING A DYING STAR — AND THE SYSTEM BUILT TO SAVE IT

MISSION CREW

Team Name · [Crack.it1914]

Crew · [Member 1, Member 2, Member 3, Member 4]

Theme · Space Systems & Crisis Response

Twelve colonies. One failing star.

Lyra Prime is dying. Around its cooling light drifts the **Helix Concord** — twelve terraformed colonies linked by a single fragile transit ring. Crops grow under artificial domes. Power is rationed. Every colony depends on the others for water, ore, and medicine.

As Lyra Prime destabilizes, it throws violent **Solar Flux Storms** across the ring — knocking out comms, frying shield grids, and stranding supply ships between colonies with no warning at all.

12

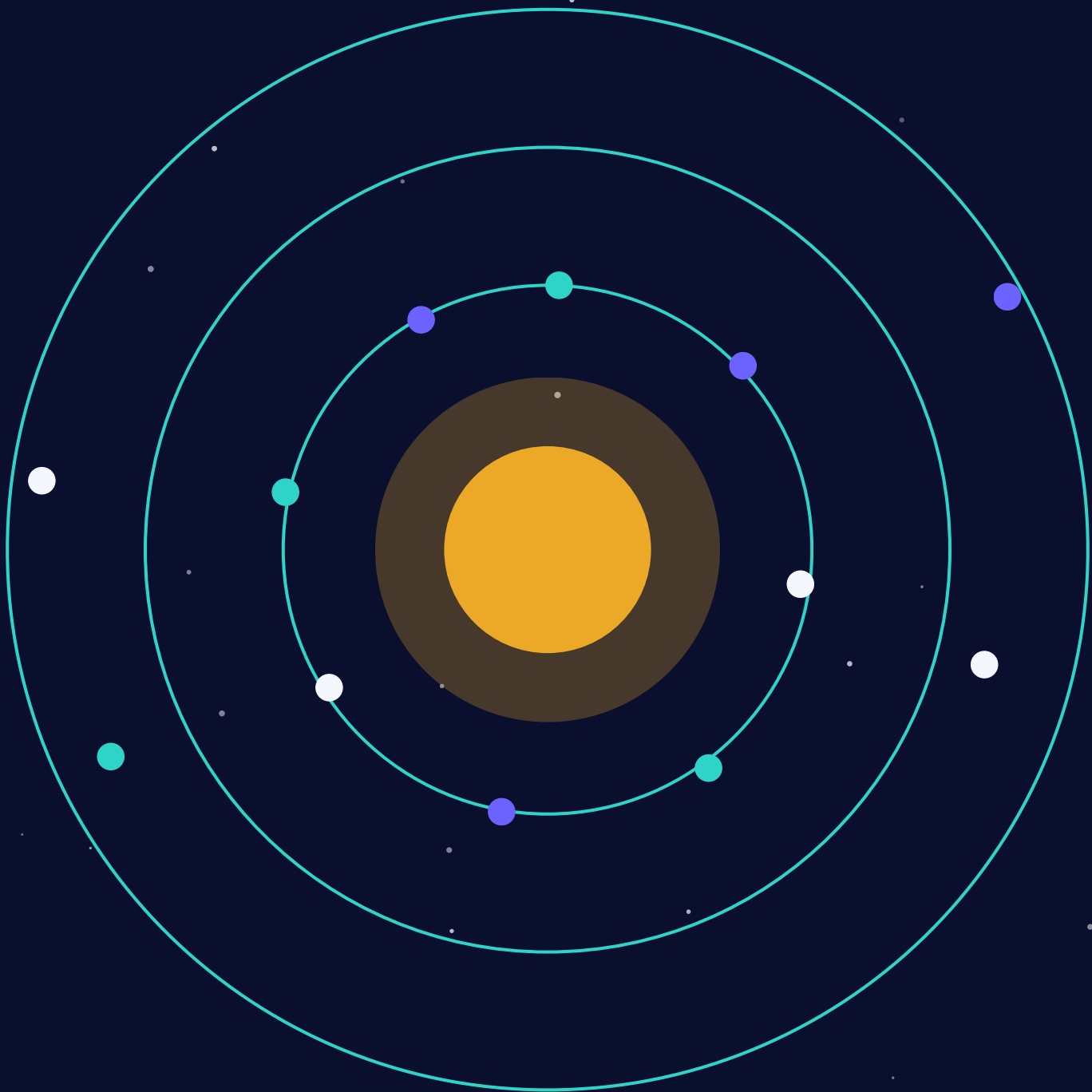
colonies on
the Helix Concord

1

shared transit ring
— zero redundancy

6 min

average warning time
before a flux storm hits



THE HELIX CONCORD

Project ASTRA

Adaptive Stellar Threat Response Array

The Concord Council has one mandate: give every colony enough warning to survive the next flux storm — and enough coordination to never do it alone.

1

Detect

Read Lyra Prime's magnetic behaviour before a storm forms, not after it arrives.

2

Predict

Forecast which colonies will be hit, how hard, and when — with enough lead time to act.

3

Coordinate

Reroute ships, share power, and trigger shield grids across colonies automatically.

MISSION CARD



THREAT

Class-IV Solar Flux Storm

STAKES

Crop domes offline, comms blackout, ships stranded mid-transit.

CURRENT WARNING TIME

6 minutes

ASTRA TARGET

40+ minutes

STATUS: MISSION ACTIVE

Sense. Predict. Act.

ASTRA is the nervous system of the Helix Concord — three linked layers turning raw stellar noise into coordinated colony action.

1 SENSE

Magnetometer buoys and colony sensors stream real-time solar activity into a shared ingestion layer.

IoT sensors · streaming pipeline

2 PREDICT

A time-series ML model forecasts storm class, path, and arrival window per colony, retrained on every event.

ML forecasting model · anomaly detection

3 ACT

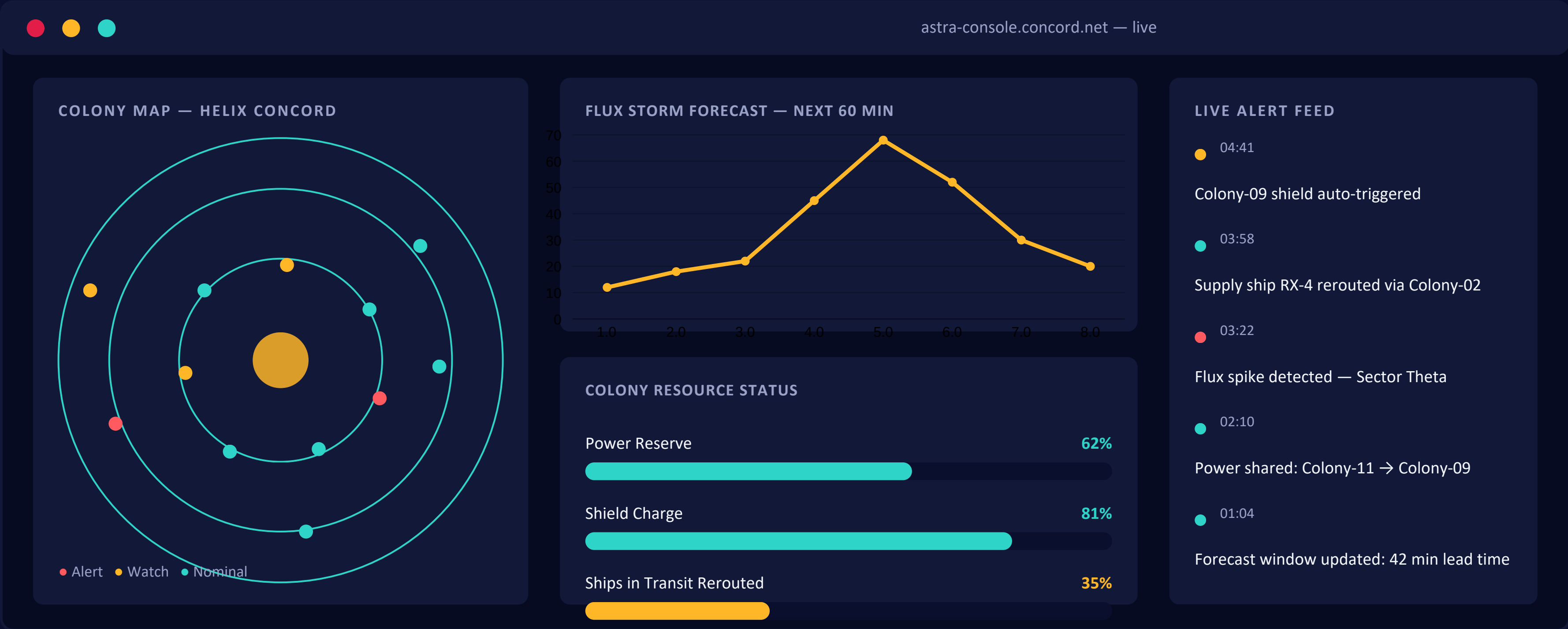
The console auto-issues shield triggers, reroutes ships, and rebalances power the moment risk crosses threshold.

Automated alerts · orchestration rules

Why it's different: most storm-alert systems stop at detection. ASTRA closes the loop — it also acts, colony by colony, in the same minute it predicts.

ASTRA Command Console

What a Concord operator sees in the minutes before a storm.





**The star is dying.
The Concord doesn't have to.**

Project ASTRA — Sense. Predict. Act.

Team Name · [Your Team Name] Crew · [Member 1, Member 2, Member 3, Member 4]

THANK YOU · QUESTIONS WELCOME